

Highway Substrate Foreign Asset Transfers Audit Summary

Prepared by Cyfrin

Version 2.0

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November 3, 2026

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Highway on Substrate Foreign Asset Transfers. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

Technical details of individual findings are omitted from this public summary to limit disclosure of security-sensitive information. The complete audit report was delivered privately to Highway, who retain it as the full record of the engagement. Recipients of the full report can use the SHA-256 hash in this summary's attestation to confirm that their copy matches the referenced report.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 3 days, the Cyfrin team conducted an audit on the Substrate Foreign Asset Transfers code provided by Highway. The findings consist of 17 Low severity issues with the remainder being informational.

Summary

Project Name	Substrate Foreign Asset Transfers
Repository	hway-substrate
Commit	e78de82f0055...
Fix Commit	cfb16b6241ad...
Audit Timeline	Oct 7th - Oct 9th, 2026
Methods	Manual Review, Proprietary AI

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	17	6	11
Informational	4	1	3
Gas Optimizations	0	0	0
Total	21	7	14

How to read the statuses

Resolved means Cyfrin verified the remediation documented in the full report with a Cyfrin verification note. Acknowledged means Highway chose to retain the reported behavior or accept the residual risk; this does not constitute resolution or endorsement by Cyfrin. Finding-specific explanations and verification notes are recorded in the full report.

6 Audit Scope

The audit scope was limited to the following files:

```
pallets/highway-config/src/lib.rs
pallets/highway-entry/src/lib.rs
runtime/src/configs/mod.rs
runtime/src/lib.rs
runtime/src/migrations.rs
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Highway (the Mosaic Bridge) carries messages and fungible tokens between the Mosaic parachain and its counterpart chains. This review concerns the Substrate implementation of foreign-asset transfers: a registered Highway token can now identify an asset by its XCM `Location`, allowing the bridge to handle both local assets and assets represented by the runtime's `ForeignAssets` pallet. The five-file scope covers bridge configuration, message entry and execution, runtime asset routing, and the migration of existing token registrations.

8.1 Scoped Components

Component	Role
`highway-config`	Stores supported chains and token registrations, per-corridor transfer modes and amount limits, decimal settings, fees, pause state, and the payload-call whitelist.
`highway-entry`	Accepts outbound messages, collects fees, burns or escrows tokens, verifies inbound relayer attestations, prevents replay, and mints or releases tokens and executes approved payloads.
Runtime asset configuration	Routes canonical local-asset `Location` IDs to `Assets` and other `Location` IDs to `ForeignAssets`. Native currency uses `Balances`; fee assets retain numeric IDs in `Assets`.
Runtime migration	Converts legacy numeric asset IDs in Highway token registrations to local-asset `Location` IDs during the runtime upgrade.

8.2 Message Lifecycle

1. **Outbound.** A user calls `emit_message` for an active destination chain with a token transfer, a payload, or both. Highway validates the configured corridor and transfer limits and verifies a signed fee quote when fee signing is enabled. It derives a domain-separated message ID, collects the fee, burns the tokens or transfers them into the configured escrow account, and emits `MessageEmitted` for off-chain relayers.
2. **Attestation.** Relayers observe the source event and produce an aggregate BLS attestation. The destination selects a committee from the registered active set for the proof's epoch; the audited runtime requires at least 86 signatures from a 128-member committee.
3. **Inbound.** An active registered relayer submits `receive_message`. The pallet reconstructs the message ID, verifies the committee attestation and replay status, then applies the current inbound corridor configuration. It converts the source amount to local precision, enforces local limits, and mints or releases the corresponding native, local, or foreign asset. An optional payload is decoded as a runtime call and dispatched from a dedicated pallet account only if its call is whitelisted.

Each corridor pairs outbound `Burn` with inbound `Mint`, or outbound `Escrow` with inbound `Release`; either direction may be closed. The relayer registry, off-chain fee and attestation services, and counterpart-chain implementations are dependencies of this flow, but are outside this review's five-file production scope.

9 Attestation

Cyfrin attests that it performed a security review of the Substrate Foreign Asset Transfers code provided by Highway.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-11-03-cyfrin-highway-substrate-foreign-asset-transfers-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Highway has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=53888e8d97bfe6542ab4779afc70b553c1d130d00614730332a7f66287c310f4
F=2026-11-03-cyfrin-highway-substrate-foreign-asset-transfers-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.

Prepared	November 3, 2026	**Authorized Lead Auditor**	Farouk Elalem
Attestation Date	November 3, 2026	**Title** / **Role**	Lead Security Researcher
Report	2026-11-03-cyfrin-highway-substrate-foreign-asset-transfers-v2.0.pdf		
Signature			